Introduction to Computer and Computer Science

Homework Assignment #12

Due: 12/26 0:00 am

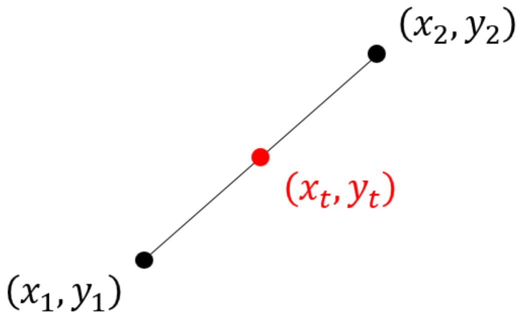
Interpolation

In order to produce additional data points, a practical method called interpolation is often applied to achieve this objective. In this homework, we are going to write a program to implement linear interpolation.

Enjoy it!

Problem #1: Linear interpolation

The working principle of linear interpolation is illustrated as below:



$$y_t = \frac{(y_2 - y_1)}{(x_2 - x_1)}(x_t - x_1) + y_1$$

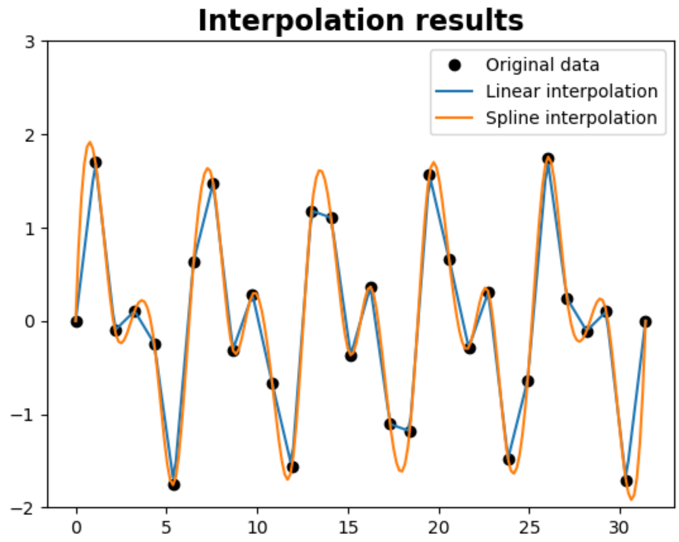
Please write an algorithm to realize this idea. For simplicity, a sample code called "**HW#12_sample code.py**" has been provided. You should put your linear interpolation algorithm into the class method called "**interp**." The data of reference points are stored in a file named "**HW#12_data.csv**."

You can test your result via the function called "**doLinearInterp**." It should return you the corresponding result.

Hint: Be aware! Please <u>do not use any open source code</u> for linear interpolation to accomplish this problem.

Problem #2: Other interpolation algorithms

In this problem, you are allowed to apply any open source code to do this interpolation task. Please find an open source code and learn how to use it. Please plot your result as shown below. The result might be different because of the interpolation method.



Please be aware, the teaching assistants will use "import" command to import your program and execute some testing code to examine your program. **The name of your functions, class object, including the name of corresponding methods, should be exactly as same as the name we shown here**. Otherwise it will pop out an error and terminate the testing procedure. In that case, your homework will be considered to be failed. **Please double check your program carefully**.

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```
# ===========================================================================
#
# Homework assignment #0
# Student: Your name, 你的名字
# Student ID: Your student ID, 你的學號
# Collaborators: John Doe, Jane Doe, ...
#
# ===========================================================================
# Your code goes here...


```

Please save your code as a ".py" file, where the file name should follow this format:

108A_hw#?_YourID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature**. Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the "New e3" system. There will be a section for uploading your homework.